Echoes of Leptis Magna

In the ancient city of Leptis Magna, a secret communication system was used to transmit encrypted messages through long stone corridors. Signals travel in straight lines and can be amplified using special echo chambers placed throughout the city.

Over time, parts of this system have fallen into disrepair, and some receivers no longer get the signal. Your task is to restore the system by placing echo chambers optimally. The city is represented as a grid of `N` rows and `M` columns. Each cell is one of the following:

- `S` — A signal source. It emits signals in all four directions (up, down, left, right).
- `E` — An echo chamber. A signal entering this cell from any direction is re-emitted in all four directions.
- `R` — A receiver. It must receive at least one signal.
- `#` — A blocked structure. Signals cannot pass through this cell.
- `.` — An empty cell where signals can pass.

A signal travels in a straight line until:

- It leaves the grid, or
- It hits a blocked structure (`#`)

Signals pass through receivers normally; only blocked structures stop them. You may convert any empty cell (`.`) into an echo chamber (`E`).

Find the **minimum number of empty cells** that must be converted into echo chambers so that **all receivers receive at least one signal**. If it is impossible, output `-1`.

Constraints

- $N, M \leq 5000$
- There is at least one source
- There is one or two recievers only

Implementation details

You must implement the following function:

```
int min_echo(int N, int M, vector<string> grid)
```

This procedure will be called exactly once. It must return the minimum number of empty cells that need to be replaced with echo chambers. If impossible, return `-1`.

Subtasks

Test group	Points	Constraints
1	5	There are no blocked structures and no echo chambers, $N, M \leq 1500$
2	8	There are no blocked structures, $N, M \leq 1500$
3	9	The solution must only determine if the answer is `-1` or not (print `0` in this case), $N, M \leq 1500$
4	14	The solution only needs to check whether the answer is `0` or not (print `-1` in this case), $N, M \leq 1500$
5	31	There is exactly one receiver, $N, M \leq 1500$
6	21	There are exactly two receivers, $N, M \leq 1500$
7	12	No additional constraints

Examples

```
min_echo(3, 5, ["S.#.R", ".S#R.", "E...."])
```

The testcase is

```
S.#.R
.S#R.
E....
```

The answer is **2**. One such ways to add echo chambers is as follows.

```
S.#ER
.S#R.
E..E.
```

Another possible configuration is

```
S.#.R
.S#R.
E..EE
```

```
min_echo(3, 5, ["S.#.R", "R####", "E..SS")
```

The testcase is

```
S.#.R
R####
E..SS
```

One of the receivers is completely blocked. The answer is **-1**

Sample Grader

The sample grader reads the input data as follows:

- Line 1: $N M$
- Line $2 + i$: $s[i]$

The sample grader prints the return values of the procedure `min_echo`.

Limits

- **Time limit:** 5 seconds
- **Memory limit:** 2048 MB